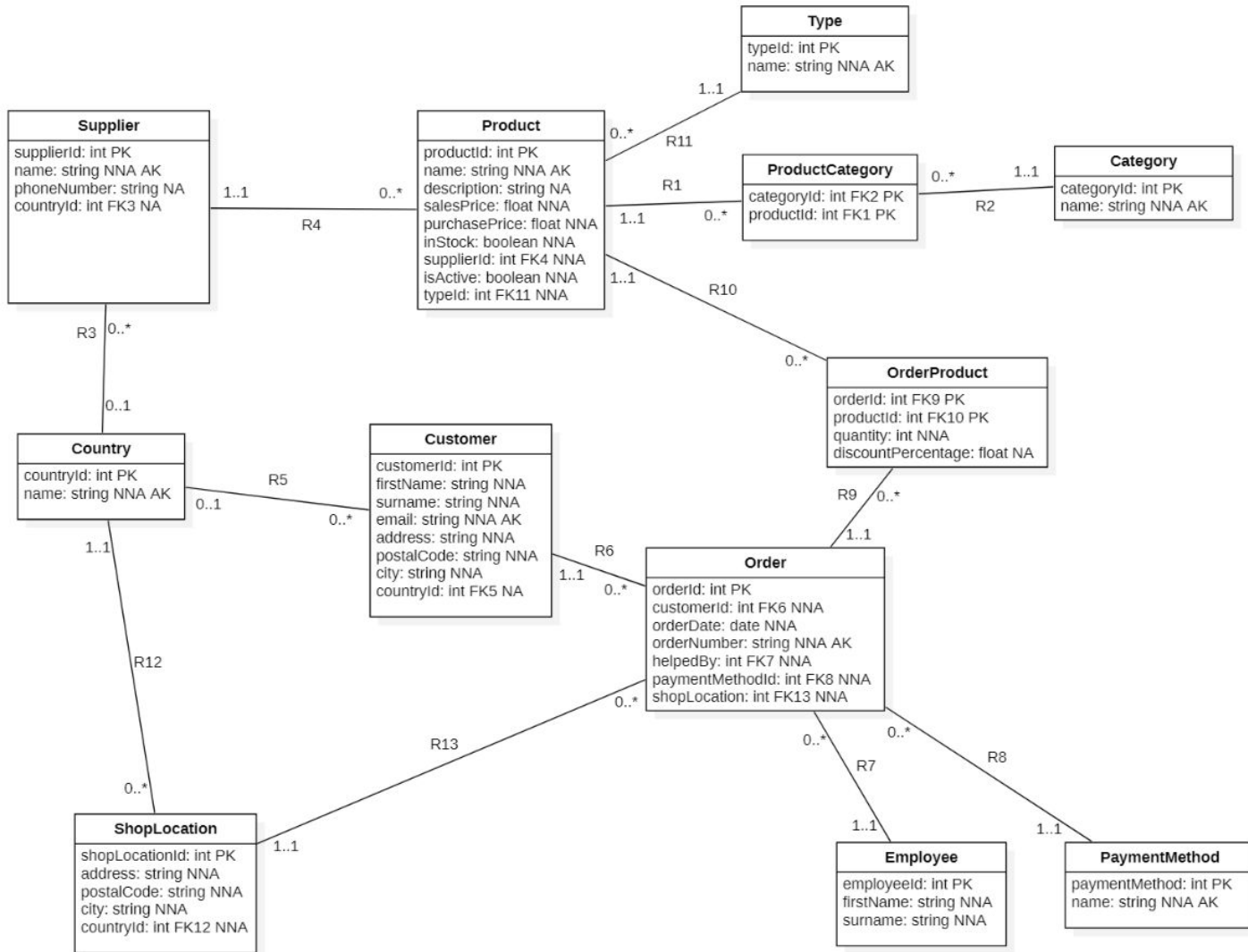


Who are we



Installation

What will we cover?



Structure of a data model

Exercise 4: Sponsorship agency

You have an agency seeking sponsorship for client projects.

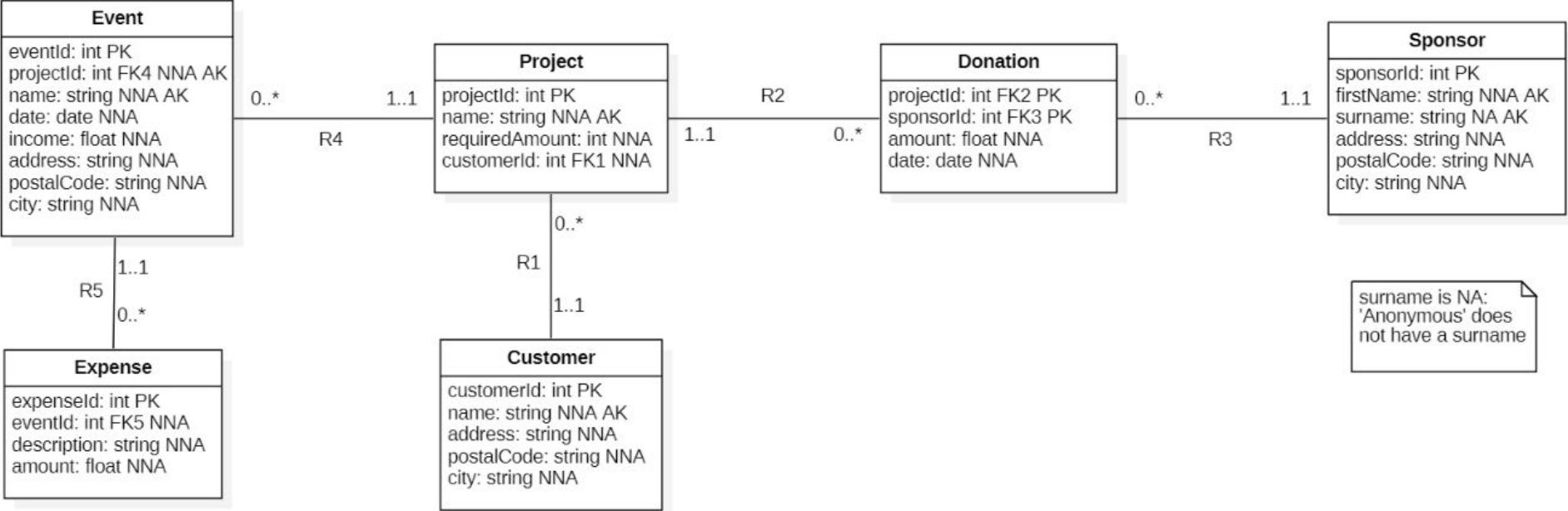
You will design your own database to process your client data with their projects and make sure you can print a record of all the sponsorship funds you have collected.

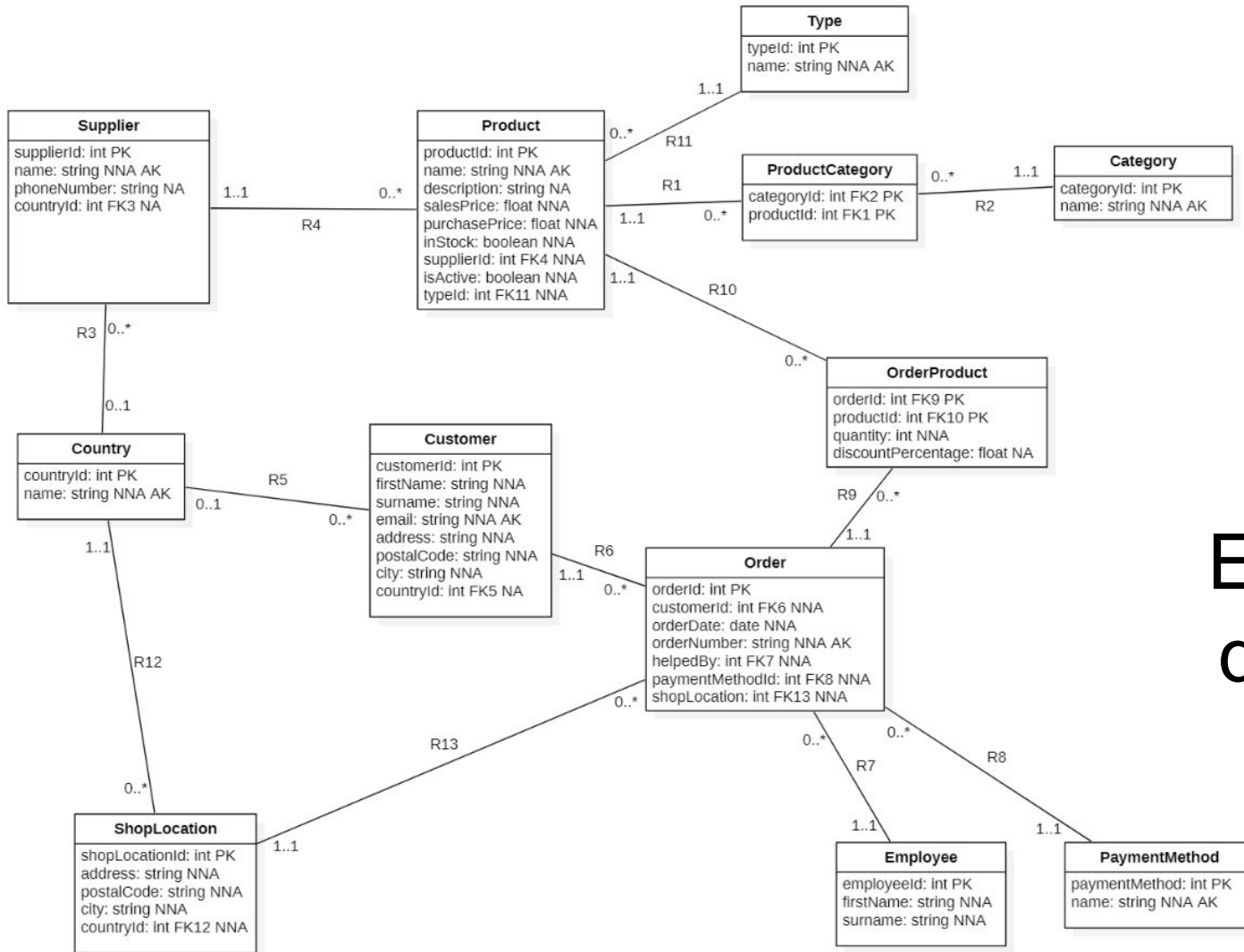
All anonymous donations are placed under the person 'anonymous'. This allows a person (e.g., "anonymous") to deposit sponsorship money multiple times. Individuals whose names are known can also make multiple sponsorship donations. The amount and date of each deposit of sponsorship money must always be recorded.

You also organize events in consultation with the client to gain sponsorship. However, there are costs associated with such events. So for each event, you should be able to provide an overview of the costs incurred.

Overview Projects and Sponsorships				
Project:	Save the museum			
Client:	VZW Polder Museum			
Address:	Dorpstraat 45 2798 Lillo			
Required amount:	100.000e			
Sponsors				
Sponsor		Amount	Date	
Peeters Marc	Namenstraat 34 2000 Antwerp	€ 100	15/01/2023	
Bleyens Robbe	Kerkstraat 45 2610 Wilrijk	€ 20	16/01/2023	
Anonymous	XXXX 0000 XXXX	€ 1.000	16/01/2023	
Anonymous	XXXX 0000 XXX	€ 150	16/01/2023	
...				
Total:		€ 24.500		
Organized events				
Event:	BBQ	Date:	20/01/2023	
Expenses				
- Hall rental		€ 150		
- Purchasing food		€ 200		
Total		€ 350		
Revenue		€ 2.300		
Event:	Ticket sales	Date:	30/01/2023	
Expenses				
- Printing tickets		€ 150		
- Gift sellers		€ 100		
Total		€ 250		
Revenue		€ 1.800		
Total events		€ 3.500		
Net revenue sponsorship		€ 28.000		
Deficit / Profit:		€ -72.000		

Interpreting
a question





Exercise on data model

Model vs SQL table

Type	Info
DATE	- Only date - Range from '1000-01-01' to '9999-12-31'
DATETIME	- Date and time - Range from '1000-01-01 00:00:00' to '9999-12-31 23:59:59'
TIMESTAMP	- Date and time - Range from '1970-01-01 00:00:001' to '2038-01-19 03:14:07'
TIME	- Time and Elapsed time - Range from -'838:59:59' to '838:59:59'

varchar
bool

Type
FLOAT(p) with 0 <= p < 24 (Storage space = 4 bytes)
FLOAT(p) with 24 <= p <= 53 (Storage space = 8 bytes)

Type	# Bytes	Min value	Max value	Signbit
TINYINT	1	-128	127	Yes
		0	255	No
SMALLINT	2	-32 768	32 767	Yes
		0	65 535	No
MEDIUMINT	3	-8 388 608	8 388 607	Yes
		0	16 777 215	No
INT	4	-2 147 483 648	2 147 483 647	Yes
		0	4 294 967 295	No
BIGINT	8	-9 223 372 036 854 775 808	9 223 372 036 854 775 807	Yes
		0	18 446 744 073 709 551 615	No

Structure of an SQL query

```
SELECT captainId  
FROM team;
```

```
SELECT playerId, name, birthdate  
FROM player  
WHERE city = 'Geel';
```

```
SELECT playerId, birthdate  
FROM player  
WHERE birthdate BETWEEN '2000-01-01' AND '2001-12-31';
```

```
SELECT playerId, name, birthdate  
FROM player  
WHERE city = 'Lichtaart'  
AND gender = 'V';
```

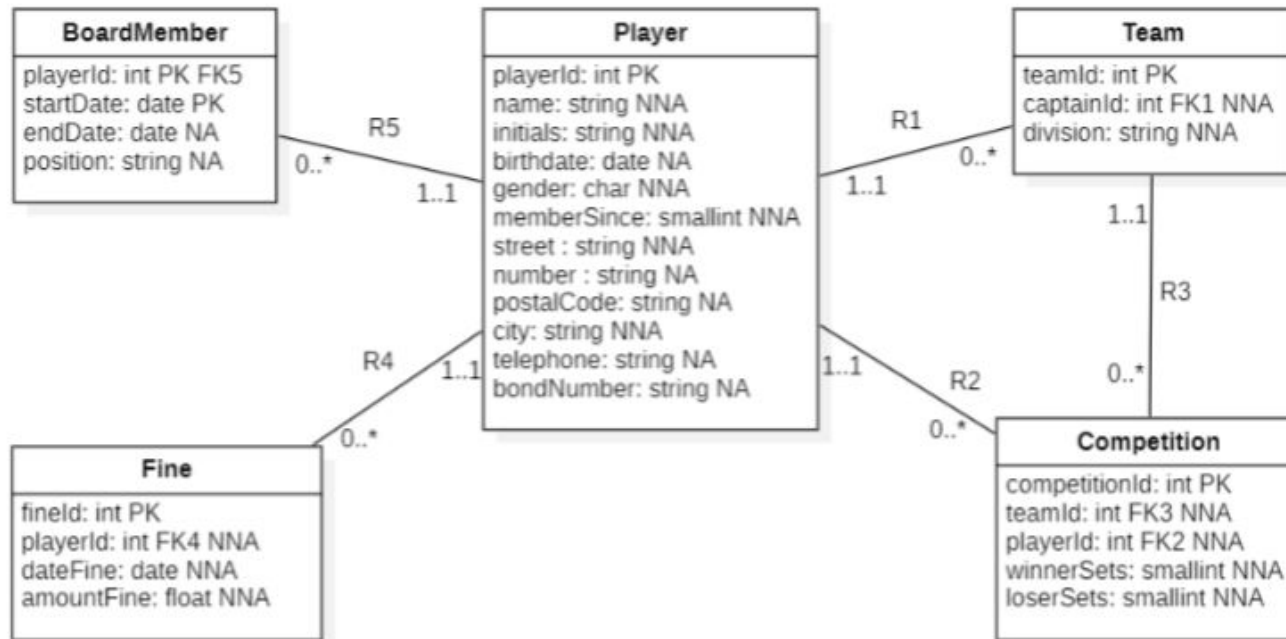
```
SELECT name  
FROM player  
WHERE name LIKE '%#_%' ESCAPE '#';
```

```
SELECT name, street, city  
FROM player  
WHERE (city = 'Geel' AND street = 'Kleinhoefstraat')  
OR (city = 'Kasterlee' AND street = 'Kastanjelaan');
```

```
SELECT playerId, bondNumber  
FROM player  
WHERE bondNumber <> '8467'  
OR bondNumber IS NULL;
```

```
SELECT playerId, name, city  
FROM player  
WHERE city IN ('Geel', 'Kasterlee', 'Lille', 'Lichtaart');
```

```
SELECT fineId, playerId  
FROM fine  
ORDER BY playerId;
```



SQL query exercise

Show the number, name and initials of each player whose name contains the letter 'a' at the second and second last position. (1 result)

	playerId	name	initials
►	112	Balen, van	J

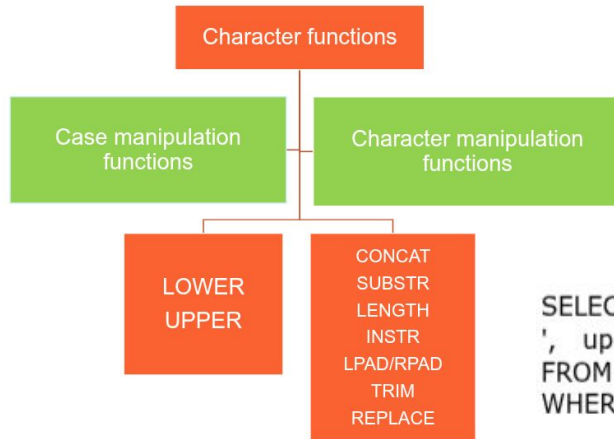
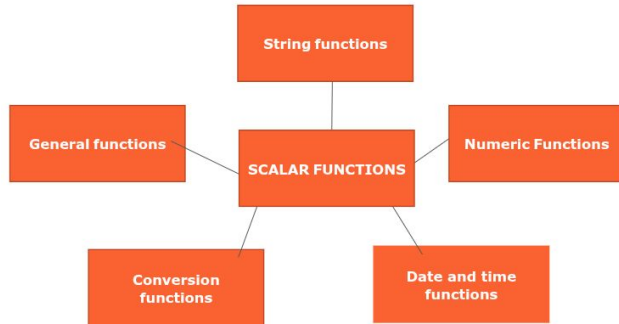
```

SELECT
FROM
WHERE
  
```

```

playerId, name, initials
tennis.player
name LIKE '_a%a_';
  
```

Scalar functions



Show an overview of all match players in the following way: "Player 'Engelen' was born on 01/09/1997, lives in GEEL and has bondnumber 2411". (10 results)

matchplayer
Player 'Engelen' was born on 01/09/1997, lives in GEEL and has bondnumber 2411
Player 'Peeters' was born on 25/06/1998, lives in GEEL and has bondnumber 8467
Player 'Nagels' was born on 08/07/2000, lives in KASTERLEE and has bondnumber 2983
Player 'Cools' was born on 28/12/1999, lives in LICHTAART and has bondnumber 2513
Player 'Bakker, de' was born on 09/01/1998, lives in KASTERLEE and has bondnumber 1124
Player 'Gorp, van' was born on 17/08/2001, lives in GEEL and has bondnumber 6409
Player 'Henderickx' was born on 11/11/2001, lives in GEEL and has bondnumber 1608
Player 'Peeters' was born on 28/02/1998, lives in GEEL and has bondnumber 6524
Player 'Mertens' was born on 10/05/2000, lives in LICHTAART and has bondnumber 7060
Player 'Balen, van' was born on 01/10/1998, lives in MOL and has bondnumber 1319

<https://dev.mysql.com/doc/refman/8.0/en/date-and-time-functions.html>

```
SELECT concat('Player "', name, '" was born on ', DATE_FORMAT(birthdate,'%d/%m/%Y'), ', ', lives in ', upper(city), ' and has bondnumber ', bondNumber) as matchplayer
FROM tennis.player
WHERE bondNumber is not null;
```

Set functions

What is the total amount of fines paid for players from Lichtaart? Show at the same time the number of different players from Lichtaart who have been fined.

	amount	number
▶	225.00	2

COUNT(*) or COUNT(playerId)

MIN(numeric value)

MAX(numeric value)

AVG(numeric value)

SUM(numeric value)

```
SELECT SUM(amountFine) as amount, COUNT(DISTINCT playerId) number
FROM   Tennis.fine
WHERE  playerId IN (SELECT playerId
                    FROM   Tennis.player
                    WHERE  city = 'Lichtaart');
```

Next week